

Microsoft Sentinel SIEM

AI-Powered Security Operations Centre

Project Implementation Documentation

Cloud Security | SIEM | SOAR | AI Integration

2026

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1 Project Overview

This document describes the end-to-end implementation of a cloud-native Security Information and Event Management (SIEM) solution using Microsoft Sentinel on Microsoft Azure, enriched with AI-powered automated incident response via OpenAI GPT. The project establishes real-time threat detection, behavioural analytics, threat intelligence watchlists, and fully automated playbook responses — all within the Azure free-tier constraints.

1.1 Goals

- Deploy a production-grade SIEM environment on Azure at zero cost using the free-tier offering.
- Ingest and analyse up to 10 GB of security logs per day.
- Detect threats using built-in analytics rules and behaviour-based UEBA.
- Block known malicious traffic sources (Tor exit nodes) via a Watchlist.
- Automate incident investigation and first-response actions using OpenAI GPT and Azure Logic Apps.

1.2 Technology Stack

Component	Service	Purpose
SIEM Platform	Microsoft Sentinel	Central log collection, analytics, and incident management
Log Storage	Log Analytics Workspace	Stores and queries ingested security data
SOAR / Automation	Azure Logic Apps	Automated playbooks triggered on incidents
AI Engine	OpenAI GPT (via API)	Generates natural-language incident analysis
Threat Intel	Watchlist (CSV)	Tor exit-node IP block list
Identity Analytics	UEBA	Behavioural baselines and anomaly detection

2 Architecture & Prerequisites

2.1 High-Level Architecture

Data sources (Azure AD, VMs, network flows, etc.) forward events to a Log Analytics Workspace. Microsoft Sentinel sits on top of this workspace and applies Analytics Rules, Watchlists, and UEBA models. When a rule fires, an Automation Rule triggers an Azure Logic App, which calls the OpenAI GPT API, collects an AI-generated analysis, and posts the result as a comment on the Sentinel incident.

2.2 Prerequisites

- An active Microsoft Azure account (free-tier is sufficient).
- Contributor or Owner role on the Azure subscription.
- An OpenAI account with an active API key (<https://platform.openai.com>).
- A GitHub account (to access the Azure-Sentinel repository).
- A modern web browser with internet access.

NOTE The Azure free account provides \$200 credit for 30 days and always-free tiers, including 10 GB/day log ingestion for Microsoft Sentinel under the free trial workspace.

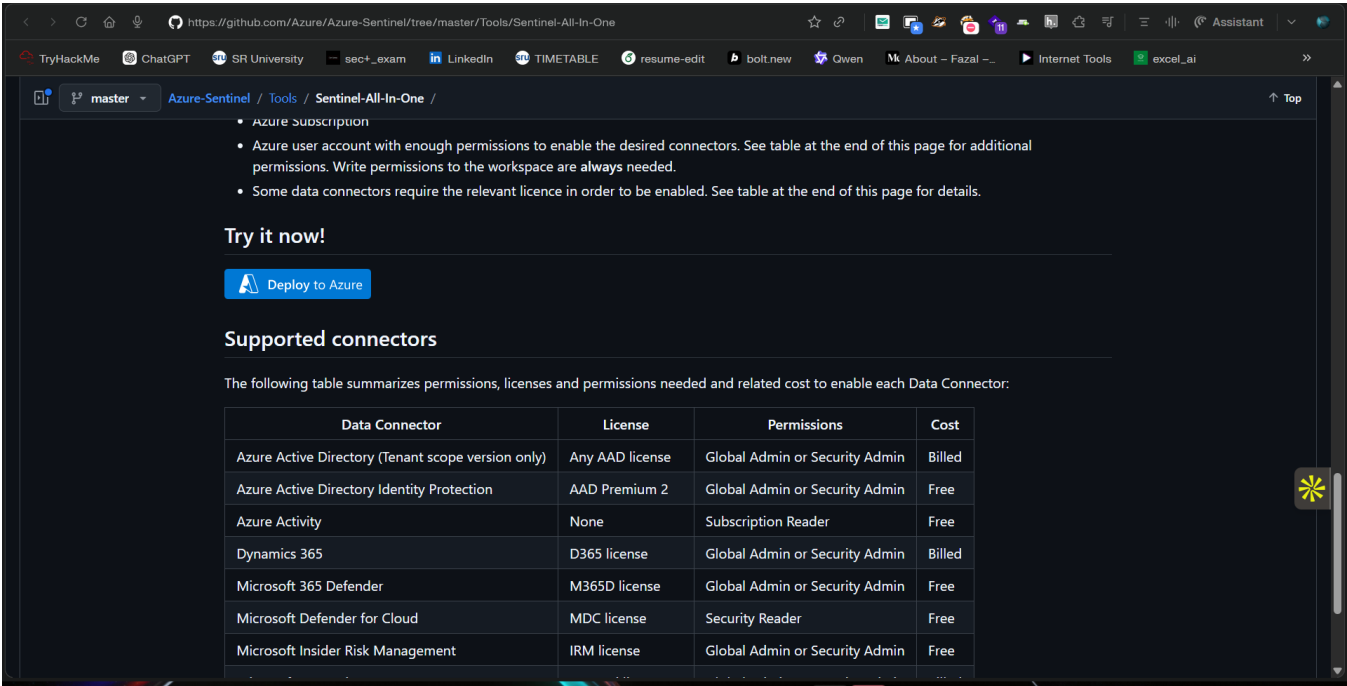
3 Deploying Microsoft Sentinel (All-In-One)

The Sentinel All-In-One template from the Azure Sentinel GitHub repository provisions the Log Analytics Workspace, installs Microsoft Sentinel, and deploys the chosen Content Hub solutions in a single ARM deployment.

3.1 Deploy from GitHub

1. Open the repository at:

<https://github.com/Azure/Azure-Sentinel/tree/master/Tools/Sentinel-All-In-One>



2. Click the Deploy to Azure button. The Azure Portal will open the custom deployment blade.
3. Sign in with your Azure credentials. If you do not have an account, click Create one and follow the free-account registration flow.

3.2 Sentinel Configuration Parameters

On the deployment form, configure the following settings:

Parameter	Recommended Value	Notes
Subscription	Your free-tier subscription	Select the subscription linked to your free account
Resource Group	Create new (e.g., rg-sentinel-lab)	Used in later steps — note the exact name
Region / Location	Nearest to your production workloads	Lower latency, lower egress costs

Workspace Name	e.g., law-sentinel-lab	Note this name — it is referenced throughout the guide
Daily Ingestion Cap	10 GB	Free-tier maximum

Home

Custom deployment

Deploy from a custom template

New! Deployment Stacks let you manage the lifecycle of your deployments. Try it now →

Project details

Deploying templates at subscription scope enables scenarios like applying policies and assigning roles at the subscription level. Subscription scope deployments are also used for creating resource groups and deploying resources in it. You can change the deployment scope by updating the schema in the template.

Subscription *

Instance details

Location *

Resource Group name *

Workspace Name *

Daily ingestion limit in GBs. Enter 0 for no limit. *

Number of days of retention *

Select pricing tier for Sentinel and Log Analytics *

Previous Next Review + create

3.3 Settings Tab

- Navigate to the Settings tab of the deployment form.
- Enable the toggle: Enable Sentinel health diagnostics.

NOTE Health diagnostics provide visibility into data-connector failures and workspace performance issues.

Home

Custom deployment

Deploy from a custom template

New! Deployment Stacks let you manage the lifecycle of your deployments. Try it now →

Basics **Settings** Content Hub Solutions Data connectors Analytics Rules Review + create

Please select which settings you would like to enable in your Microsoft Sentinel environment

Enable User Entity Behavior Analytics (UEBA)? Global Admin or Security Admin permission required. ☐

Enable Sentinel health diagnostics? ☒

Previous Next Review + create

3.4 Content Hub Solutions Tab

- Open the Content Hub Solutions tab.
- Click Select All for each of the three option groups (Microsoft Solutions, Partner Solutions, Community Solutions).

NOTE This installs all available detection packages, workbooks, and playbook templates. Individual solutions can be disabled later if not required.

The screenshot shows the 'Custom deployment' page in the Microsoft Sentinel portal, specifically the 'Content Hub Solutions' tab. The left sidebar contains navigation links for 'Create a resource', 'Home', 'Dashboard', 'All services', and 'FAVORITES' (Application Templates, All resources, Resource groups, App Services, Function App, Azure SQL Database, Azure Cosmos DB, Virtual machines, Load balancers, Storage accounts, Virtual networks, Microsoft Entra ID, Monitor, Advisor, Microsoft Defender for Cloud). The main content area has tabs for 'Basics', 'Settings', 'Content Hub Solutions' (active), 'Data connectors', 'Analytics Rules', and 'Review + create'. A message states: 'New! Deployment Stacks let you manage the lifecycle of your deployments. Try it now →'. Below this, a blue box says: 'Please select which Solutions you want to install into your Microsoft Sentinel environment. Solutions may contain different content types, like data connectors, analytics rules, parsers, workbooks, playbooks, hunting queries, etc.' Three selection boxes are present: 'Select Microsoft Content Hub solutions to install' (12 selected), 'Select Essentials Content Hub solutions to install' (9 selected), and 'Select Training and Tutorials Content Hub solutions to install' (2 selected). At the bottom are 'Previous', 'Next', and 'Review + create' buttons.

3.5 Data Connectors Tab

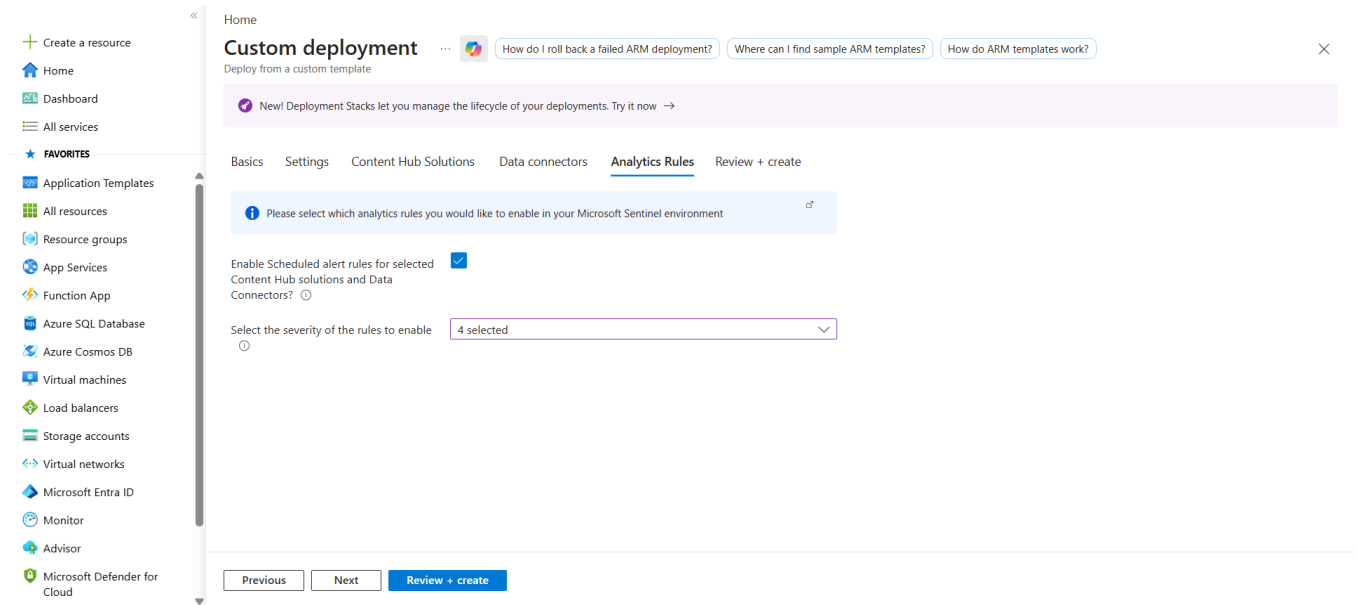
8. Open the Data Connectors tab.
9. Click Select All to enable every available connector.

NOTE Some connectors require additional licences (e.g., Microsoft 365 Defender). Unlicensed connectors will produce errors during deployment — this is expected and handled in Section 4.

The screenshot shows the 'Custom deployment' page in the Microsoft Sentinel portal, specifically the 'Data connectors' tab. The left sidebar is identical to the previous screenshot. The main content area has tabs for 'Basics', 'Settings', 'Content Hub Solutions', 'Data connectors' (active), 'Analytics Rules', and 'Review + create'. A yellow warning box states: 'Changes on this step may reset later selections you have made. Review all options prior to deployment.' Below this, a blue box says: 'Please select which Data Connectors you would like to configure in your Microsoft Sentinel environment'. Two selection boxes are present: 'Select data connectors to onboard *' (11 selected) and 'Select Azure Active Directory log types to enable' (12 selected). At the bottom are 'Previous', 'Next', and 'Review + create' buttons.

3.6 Analytics Tab

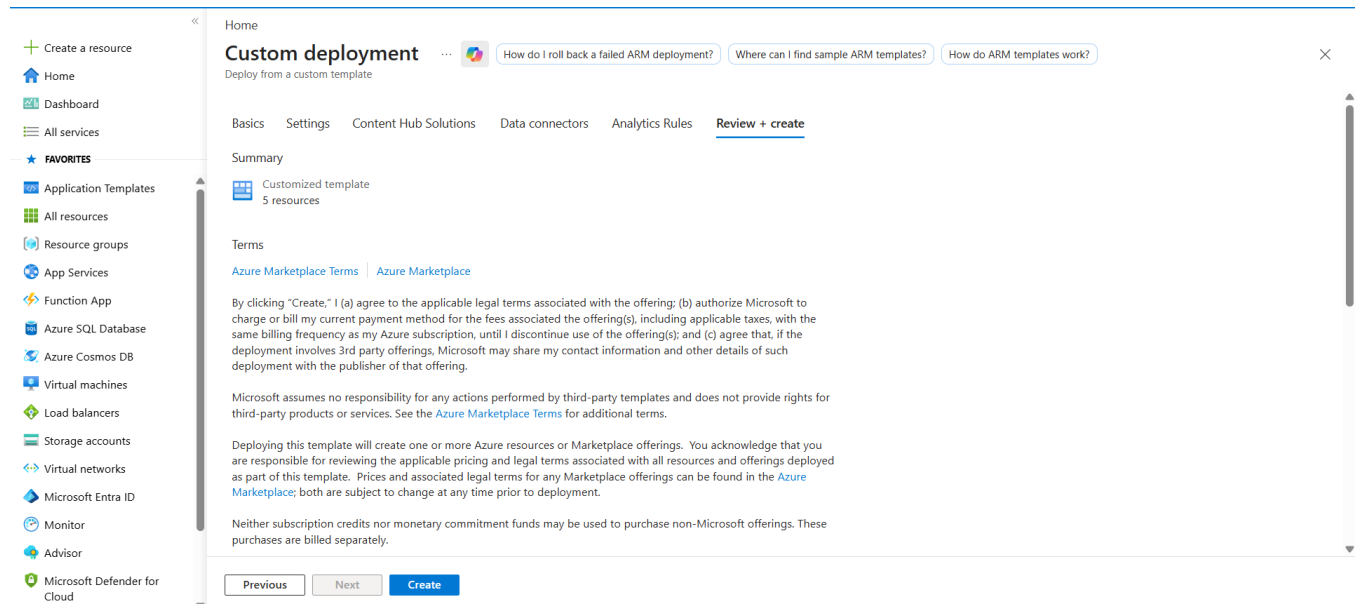
10. Open the Analytics tab.
11. Tick the top-level checkbox to select all analytics rules templates.
12. Confirm that all rules in the list below are also selected.



3.7 Review + Create

13. Click Review + Create.
14. Scroll to the bottom and click Create.
15. The deployment will run for approximately 10 to 15 minutes.
16. A partial failure related to data connectors is expected — it will auto-resolve. Proceed to Section 4.

NOTE If the overall deployment shows as Failed, do not delete the resource group. The Sentinel workspace itself is likely healthy. Only the unsupported connector resources will have failed.

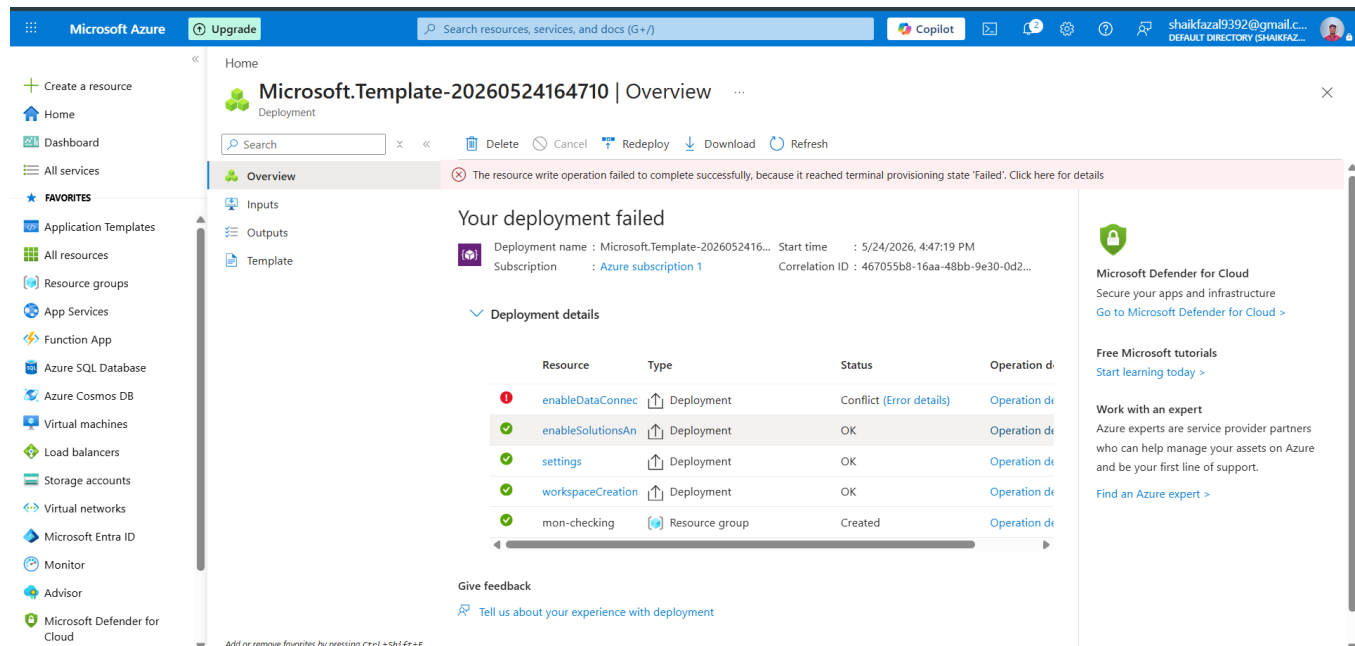


4 Sentinel Configuration

4.1 Fixing the Diagnostic Settings (Post-Deployment)

After deployment completes (even partially), configure diagnostic settings on the Log Analytics Workspace to ensure Sentinel health data flows correctly.

17. In the Azure Portal left sidebar, click Resource Groups.
18. Select the resource group you created in Step 3.2.
19. Inside the resource group, click the Log Analytics Workspace resource (the one matching your workspace name).
20. In the left menu of the workspace, scroll to the Monitoring section and click Diagnostic settings.
21. Click + Add diagnostic setting.
22. Configure the diagnostic setting as follows:
 - Name: sentinel-diagnostics (or any meaningful label)
 - Logs category: Select all log categories shown
 - Destination: Send to Log Analytics Workspace
 - Workspace: Select the same workspace (law-sentinel-lab)
23. Click Save.



Microsoft Azure | Upgrade | Search resources, services, and docs (G+/) | Copilot | shaikfaza19392@gmail.c... | DEFAULT DIRECTORY (SHAIKFAZ...)

Home | Microsoft.Template-20260524164710 | Overview | Deployment

Search | Delete | Cancel | Redeploy | Download | Refresh

Overview | The resource write operation failed to complete successfully, because it reached terminal provisioning state 'Failed'. Click here for details

Your deployment failed

Deployment name : Microsoft.Template-2026052416... Start time : 5/24/2026, 4:47:19 PM
Subscription : Azure subscription 1 Correlation ID : 467055b8-16aa-48bb-9e30-0d2...

Deployment details

Resource	Type	Status	Operation details
enableDataConnect	Deployment	Conflict (Error details)	Operation details
enableSolutionsAn	Deployment	OK	Operation details
settings	Deployment	OK	Operation details
workspaceCreation	Deployment	OK	Operation details
mon-checking	Resource group	Created	Operation details

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Find an Azure expert >

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Create a resource

Home

Dashboard

All services

FAVORITES

Application Templates

All resources

Resource groups

App Services

Function App

Azure SQL Database

Azure Cosmos DB

Virtual machines

Load balancers

Storage accounts

Virtual networks

Microsoft Entra ID

Monitor

Advisor

Microsoft Defender for Cloud

Home > mon-checking | Diagnostic settings

Diagnostic setting

Save Discard Delete Feedback

A diagnostic setting specifies a list of categories of platform logs and/or metrics that you want to collect from a resource, and one or more destinations that you would stream them to. Normal usage charges for the destination will occur. [Learn more about the different log categories and contents of those logs](#)

A more flexible, faster, and robust way to collect metrics is in preview! Click [here](#) to configure platform metrics collection from microsoft.operationalinsights/workspaces to storage account, event hubs, and Log Analytics workspace. [Learn more.](#)

Diagnostic setting name *

sentinal

Logs

Category groups

audit

allLogs

Categories

Audit

Summary Logs

Job Logs

Metrics

AllMetrics

Destination details

Send to Log Analytics workspace

Subscription

Azure subscription 1

Log Analytics workspace

mon-checking (westus)

Archive to a storage account

Stream to an event hub

Send to partner solution

Create a resource

Home

Dashboard

All services

FAVORITES

Application Templates

All resources

Resource groups

App Services

Function App

Azure SQL Database

Azure Cosmos DB

Virtual machines

Load balancers

Storage accounts

Virtual networks

Microsoft Entra ID

Monitor

Advisor

Microsoft Defender for Cloud

Home > mon-checking

mon-checking | Diagnostic settings

Log Analytics workspace

Search Refresh Feedback

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Logs

Resource visualizer

Settings

Classic

Monitoring

Insights

Alerts

Metrics

Diagnostic settings

Advisor recommendations

Workbooks

Diagnostic settings are used to configure streaming export of platform logs and metrics for a resource to the destination of your choice. You may create up to five different diagnostic settings to send different logs and metrics to independent destinations. [Learn more about diagnostic settings](#)

Diagnostic settings

Name	Storage account	Event hub	Log Analytics workspace	Partner solution	Edit setting
sentinal	-	-	mon-checking	-	Edit setting

+ Add diagnostic setting

Click 'Add Diagnostic setting' above to configure the collection of the following data:

- Audit
- Summary Logs
- Job Logs
- AllMetrics

The image shows two screenshots of the Microsoft Sentinel portal. The top screenshot displays the 'mon-checking' resource group overview, and the bottom screenshot displays the 'Microsoft Sentinel' workspace overview.

Top Screenshot: mon-checking Resource Group Overview

Home

mon-checking

Resource group

Search

Essentials

Resources

Name	Type	Location
deployRules	Deployment Script	West US
e8678543-1f93-4823-8efe-8f5c9e98b0eb (- 05/24/2026 11:23)	Azure Workbook	West US
hor74bn42hn2azscripts	Container instances	West US
hor74bn42hn2azscripts	Storage account	West US
mon-checking	Log Analytics workspace	West US
runPowerShellInline	Deployment Script	West US
SecurityInsights(mon-checking)	Solution	West US
sleep	Deployment Script	West US
useridentityq2zqpis5txwjm	Managed Identity	West US

Showing 1 - 9 of 9. Display count: auto

Bottom Screenshot: Microsoft Sentinel Workspace Overview

Home

Microsoft Sentinel

Default Directory (shaikfaza9392gmail.onmicrosoft.com)

Subscription equals all

Name	Resource Group	Location	Subscription	Directory
mon-checking	mon-checking	West US	Azure subscription 1	Default Directory

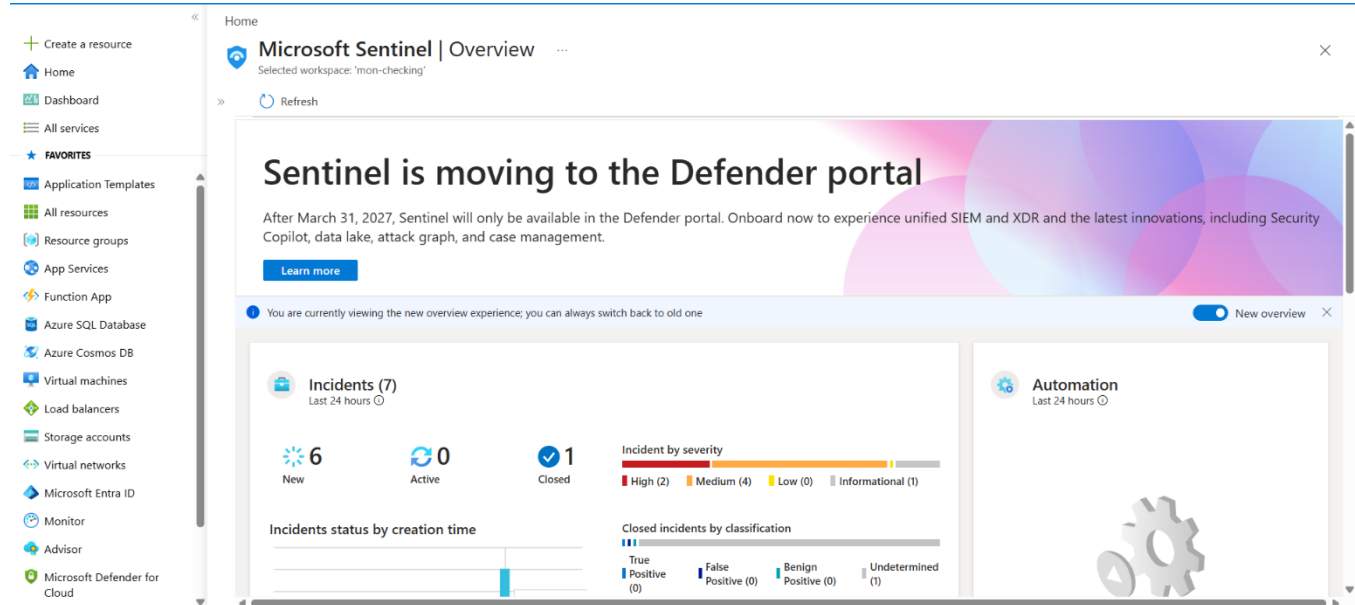
Showing 1 - 1 of 1. Display count: auto

4.2 Opening Microsoft Sentinel

24. In the Azure Portal search bar, type Microsoft Sentinel and press Enter.

25. In the Sentinel workspaces list, click the workspace name you created earlier.

You are now inside the Sentinel portal. All remaining configuration steps will take place here unless otherwise stated.



5 Enabling UEBA (Behaviour-Based Detection)

User and Entity Behaviour Analytics (UEBA) creates behavioural baselines for users, devices, and applications, and raises alerts when activity deviates from the baseline. This provides detection capability beyond signature-based rules.

26. Inside Microsoft Sentinel, click Settings in the left sidebar.

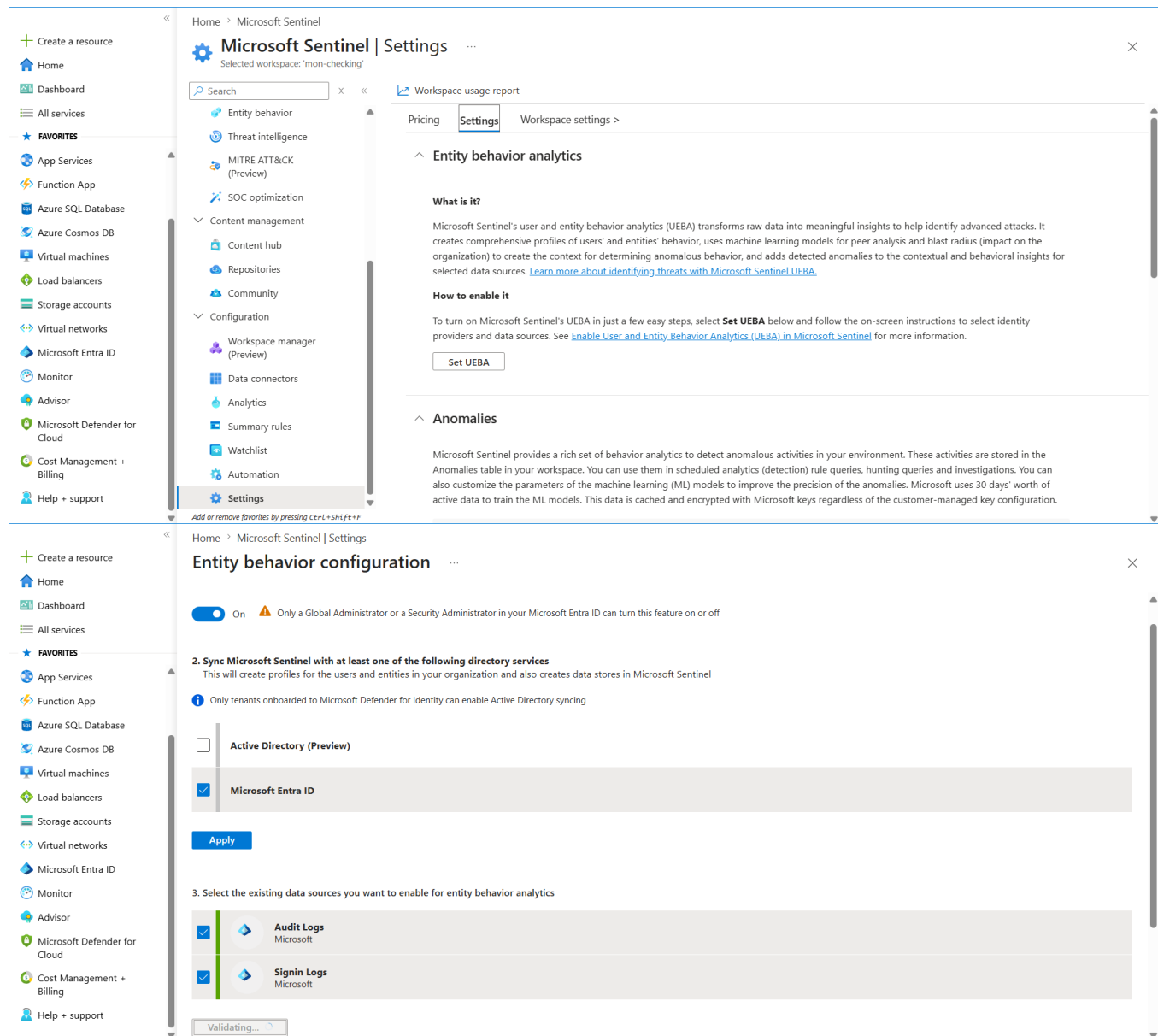
27. On the Settings page, click the Entity behavior settings section or the Set UEBA button.

28. In the UEBA configuration panel, enable the following entity types by ticking the corresponding checkboxes:

- Users
- Groups
- Hosts / Devices
- Azure resources (if applicable)

29. Click Apply to save the configuration.

NOTE UEBA may take 24 to 48 hours to accumulate enough data to generate meaningful anomaly scores after it is first enabled.



6 Playbook Permissions Setup

Sentinel must be granted permission to trigger Logic App playbooks. This is configured on the resource group level.

30. In Microsoft Sentinel, click Settings in the left sidebar.
31. Scroll down to find the Playbook permissions section.
32. Click Configure permissions.
33. In the panel that opens, locate and select your resource group (e.g., rg-sentinel-lab).
34. Click Apply.

Sentinel now has the authority to execute Logic Apps residing in that resource group when an automation rule fires.

Manage permissions

Choose the resource groups that contain the playbooks you want to give Microsoft Sentinel permissions to run

Browse **Current permissions**

Search

Name ↑↓	Subscription ↑↓
☒ mon-checking	Azure subscription 1
☐ NetworkWatcherRG	Azure subscription 1
☐ RG-honeypot	Azure subscription 1

Apply **Cancel**

7 Watchlist Setup — Tor Exit Nodes

A Watchlist imports external threat intelligence into Sentinel in CSV format, making it queryable inside KQL analytics rules. This project uses a Tor exit-node IP list to detect and alert on connections from anonymisation infrastructure.

7.1 Obtaining the CSV File

35. Open the GitHub repository:

<https://github.com/Azure/Azure-Sentinel/tree/master/Tools/Sentinel-All-In-One>

36. Browse to the Sample Data or Watchlists directory.

37. Download the Tor exit nodes CSV file to your local machine.

NOTE Alternatively, you can obtain a current list of Tor exit nodes from <https://check.torproject.org/exit-addresses> and format it as a CSV with columns: IpAddress, Description.

Microsoft Sentinel | Watchlist

Selected workspace: 'mon-checking'

Refresh New Delete Update watchlist Columns Guides & Feedback

15 Watchlists 954 Watchlist Items

My Watchlists **Templates (Preview)**

Search by name, alias and description Add filter

☐	Name	Alias	Source	Create...	Last u...
☐	D365-SecurityConfig	D365-Secs	ContentHi	5/24/2026	5/24/2026
☐	D365-UserConfig	D365-User	ContentHi	5/24/2026	5/24/2026
☐	NetworkSession Monitor Configuration	NetworkSa	NetworkSa	5/24/2026	5/24/2026
☐	SOC Email Distribution	SOCEmail	SOCEmail	5/24/2026	5/24/2026
☐	SOC General IT	SOCGener	SOCGener	5/24/2026	5/24/2026
☐	SOC Contacts	SOCConta	SOCConta	5/24/2026	5/24/2026
☐	SOC Pager Contacts	SOCPager	SOCPager	5/24/2026	5/24/2026
☐	SOC Recommended Actions	SocRA	SocRA.csv	5/24/2026	5/24/2026
☐	SOC Departmental	SOCDepar	SOCDepar	5/24/2026	5/24/2026
☐	SOC Workstations	SOCworks	SOCworks	5/24/2026	5/24/2026

No watchlist selected
Select a watchlist for more details

The screenshot shows the Microsoft Sentinel Watchlist wizard. The left sidebar contains navigation links: 'Create a resource', 'Home', 'Dashboard', 'All services', 'FAVORITES', 'Application Templates', 'All resources', 'Resource groups', 'App Services', 'Function App', 'Azure SQL Database', 'Azure Cosmos DB', 'Virtual machines', 'Load balancers', 'Storage accounts', 'Virtual networks', 'Microsoft Entra ID', 'Monitor', 'Advisor', and 'Microsoft Defender for Cloud'. The main content area is titled 'Watchlist wizard' and has three tabs: 'General', 'Source', and 'Review + create'. The 'General' tab is active, showing fields for 'Name *' (filled with 'Tor-nodes-ip'), 'Description' (empty), and 'Alias *' (filled with 'ip-nodes'). A 'Next: Source >' button is at the bottom left, and a 'Give feedback' link is at the bottom right.

7.2 Importing the Watchlist

38. In Microsoft Sentinel, locate Watchlist in the left sidebar (under Threat management) and click it.

39. Click + New to create a new watchlist.

40. Fill in the details:

- Name: TorExitNodes
- Alias: TorExitNodes (this is the name used in KQL queries)
- Description: Known Tor exit-node IP addresses

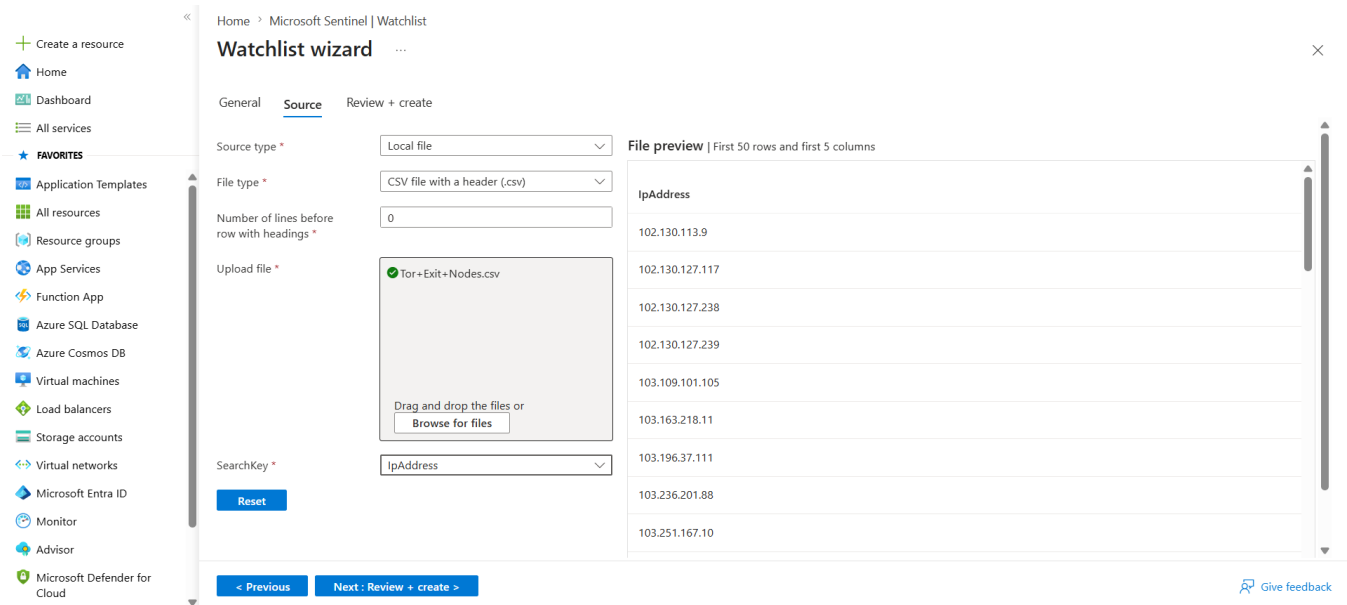
41. Under Source, click Browse and upload the CSV file downloaded in Step 7.1.

42. Set the SearchKey to IPAddress (the column used for lookups).

43. Click Review + Create and then Create.

Once imported, you can reference the watchlist in KQL analytics rules using the `_GetWatchlist()` function:

```
let TorExitNodes = (_GetWatchlist('TorExitNodes') | project IPAddress);  
SigninLogs | where IPAddress in (TorExitNodes)
```

8 Creating Analytics Rules

Analytics Rules are the detection engine of Microsoft Sentinel. Each rule runs a KQL query on a schedule and generates an alert (and optionally an incident) when the query returns results.

8.1 Types of Analytics Rules

Rule Type	Description	Best For
Scheduled	Runs a KQL query on a defined schedule (e.g., every 5 minutes) and raises alerts on matching results	Known attack patterns
Microsoft Security	Automatically creates Sentinel incidents from alerts generated by other Microsoft security products	Microsoft 365 Defender, Defender for Cloud
Fusion	Uses ML to correlate low-fidelity signals into high-confidence multi-stage attack detections	Advanced persistent threats
ML Behavioural Analytics	Applies built-in Microsoft ML models to detect anomalous activity patterns	Insider threats, compromised accounts
Anomaly	Generates anomaly signals using UEBA data, which can then be correlated in Scheduled rules	Baseline deviation detection
Threat Intelligence	Matches indicators of compromise from Threat Intelligence connectors against log data	Known IOC correlation

8.2 Creating a Scheduled Analytics Rule

44. In Microsoft Sentinel, click Analytics in the left sidebar.
45. Click + Create and select Scheduled query rule.

46. On the General tab, provide:

- Name: A descriptive name, e.g., Tor Exit Node Sign-In Detected
- Description: Detects Azure AD sign-ins originating from known Tor exit nodes
- Tactics: Map to relevant MITRE ATT&CK tactics (e.g., Initial Access)
- Severity: Medium or High, depending on your environment's risk tolerance

47. On the Set rule logic tab, enter the KQL query:

```
let TorExitNodes = (_GetWatchlist('TorExitNodes') | project IPAddress);
SigninLogs
| where IPAddress in (TorExitNodes)
| project TimeGenerated, UserPrincipalName, IPAddress, Location, ResultType
```

48. Set the Rule query runs every: 5 Minutes.

49. Set the Lookup data from the last: 5 Minutes.

50. On the Incident settings tab, enable Create incidents from alerts.

51. On the Automated response tab, you can attach the playbook created in Section 9 (do this after Section 9 is complete).

52. Click Review + Create and then Save.

8.3 Tips for Effective Rules

- Start with a broad query, validate it in Log Analytics, then narrow the scope.
- Use the Alert threshold setting to avoid alert storms from noisy data sources.
- Use Entity Mapping to link alert results to Sentinel entities (IP, User, Host) so UEBA enrichment is applied automatically.
- Review and enable the built-in Scheduled rule templates from the Rule Templates gallery — hundreds of community-maintained detections are available out of the box.

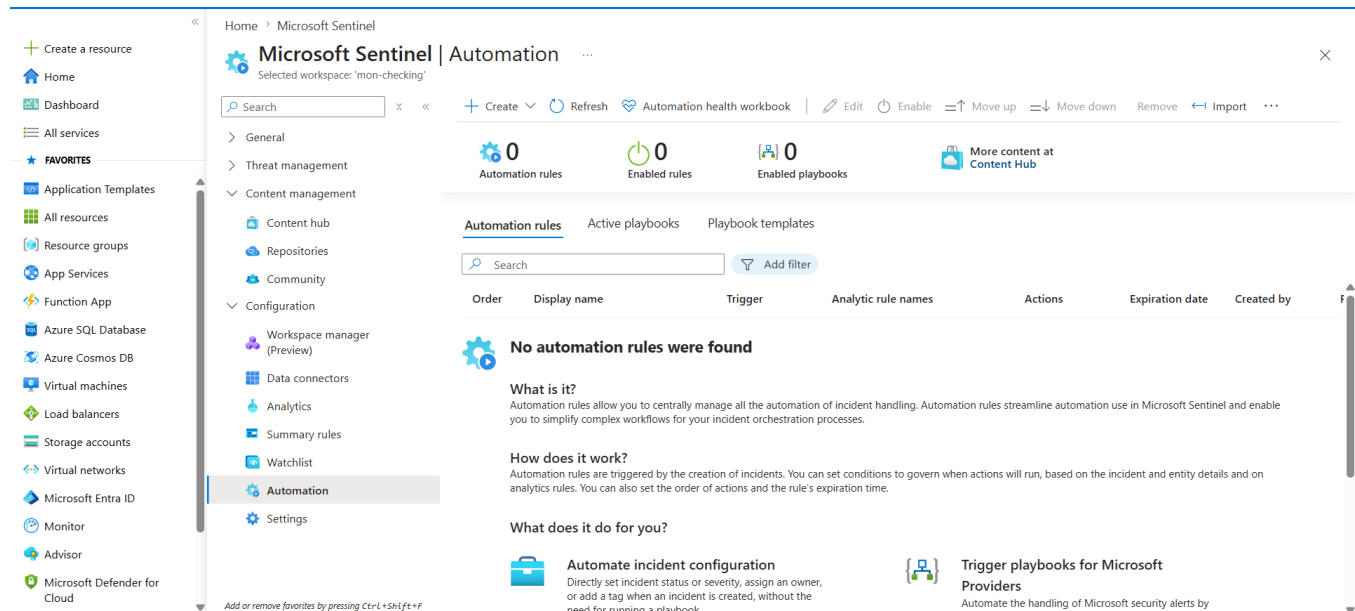
NOTE KQL queries in analytics rules are limited to a 14-day lookback. For historical trend analysis, use Workbooks instead.

9 AI-Automated Incident Response with OpenAI GPT

This section builds an Azure Logic App that acts as a Sentinel playbook. When triggered by an incident, the Logic App calls the OpenAI GPT API to generate an AI-produced incident analysis and posts the result as a comment directly on the Sentinel incident, providing an analyst with an immediate first-response recommendation.

9.1 Creating the Logic App (Playbook)

53. In Microsoft Sentinel, click Settings in the left sidebar.
54. Scroll to Playbook permissions and ensure permissions are configured (Section 6).
55. In the left sidebar click Automation, then click + Create and select Playbook.
56. Configure the playbook:
 - Subscription & Resource Group: Select the same group as your Sentinel workspace
 - Playbook name: e.g., GPT-Incident-Responder
 - Region: Same as your workspace
57. Click Review + Create, then Create.
58. Once the Logic App resource is created, click Go to resource to open the Logic App Designer.



9.2 Building the Logic App Flow

59. In the Logic App Designer, the designer opens with a Microsoft Sentinel Incident trigger already configured (When a Microsoft Sentinel Incident is created or updated). Confirm this trigger is present.
60. Click + New Step.
61. In the search box, type HTTP and select the HTTP action.
62. Configure the HTTP action as follows:
 - Method: POST
 - URI: `https://api.openai.com/v1/chat/completions`
 - Headers: Add Content-Type = `application/json` and Authorization = Bearer `<YOUR_OPENAI_API_KEY>`
 - Body: Paste the JSON payload below (modify the prompt as needed)

```
{
  "model": "gpt-3.5-turbo",
  "messages": [
    {
      "role": "system",
      "content": "You are a cybersecurity analyst assistant. Analyse the following security incident and provide a concise investigation summary, likely attack vector, recommended containment actions, and suggested next steps."
    },
    {
      "role": "user",
      "content": "Incident Title:
@{triggerBody()?['object']?['properties']?['title']}\nSeverity:
@{triggerBody()?['object']?['properties']?['severity']}\nDescription:
@{triggerBody()?['object']?['properties']?['description']}"
    }
  ],
  "max_tokens": 500
}
```

```
| }
```

63. Click + New Step again.

64. Search for Add comment to incident and select the Microsoft Sentinel action of that name.

65. Configure the Add comment to incident action:

- Incident ARM ID: Click inside the field, then in the dynamic content panel click See more and select Incident ARM ID
- Comment: Click inside the field, open dynamic content, click See more, and select the Body text field from the HTTP response. This inserts the GPT-generated analysis.

66. In the top-left of the Logic App Designer, click Save.

Home > Microsoft Sentinel | Automation > gpt-response

gpt-response | Logic app designer

Search

A new Logic Apps experience is available for preview! Try it now

Run Save Discard Parameters Code view Connections Errors Info

Create connection

GPT3 Completes your prompt

Create a new connection

Connection Name * new_conn_8fb8a

API Key * [masked]

Independent Publisher Connectors are developed and maintained by Independent Publishers. Independent Publisher Connectors are provided "as is" without warranty and are not supported by Microsoft or the underlying external service behind the connector. Only create a connection to an Independent Publisher Connector if you trust the publisher and source as the Independent Publisher Connector could introduce security or privacy risks. [Learn more](#)

By selecting **Create new** you agree to the [MIT License](#).



GPT3 Completes your prompt



Parameters

Settings

Code view

Testing

About

Engine *

DaVinci - most expensive



prompt *

How can I remediate Incident Title with description of in Incident Description

Advanced parameters

Showing 6 of 8



Show all

Clear all

n

1



best_of

1



temperature



gpt-response | Logic app designer



Logic app

A new Logic Apps experience is available for preview!

Try it now



Run

Save

Discard

Parameters

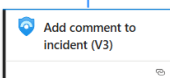
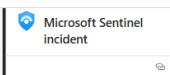
Code view

Connections

Errors

Info

▼



Add comment to incident (V3)



Parameters

Settings

Code view

Testing

About

Incident ARM id *

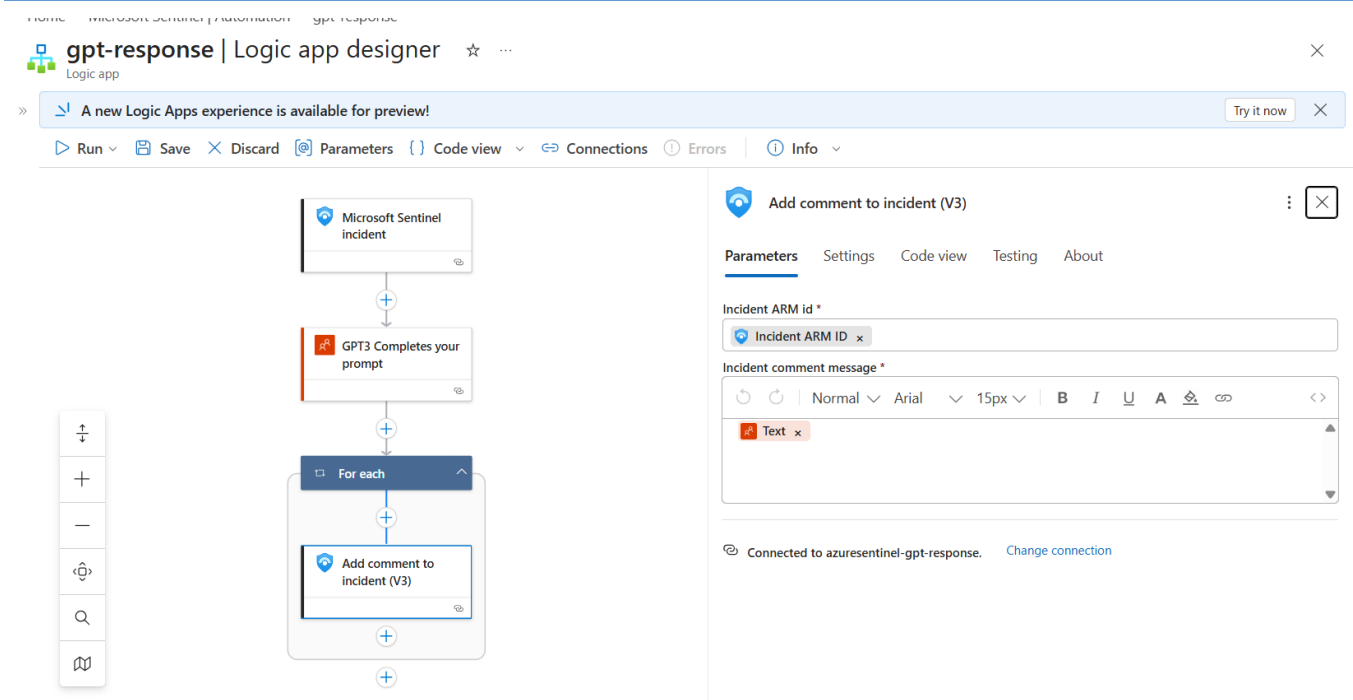
Incident ARM id

Incident comment message *

Incident comment message

Connected to azuresentinel-gpt-response.

[Change connection](#)



9.3 Obtaining an OpenAI API Key

To call the GPT API, you need an API key from OpenAI.

67. Go to <https://platform.openai.com> and sign in or create a free account.
68. In the top-right menu, click your profile avatar and select API Keys (or navigate to <https://platform.openai.com/api-keys>).
69. Click + Create new secret key.
70. Give the key a name (e.g., sentinel-playbook) and click Create secret key.
71. Copy the key immediately — it will not be shown again. Store it in a secure location such as Azure Key Vault.
72. Paste this key into the Authorization header of the HTTP action in your Logic App (prefix with Bearer — e.g., Bearer sk-...).

NOTE Free-tier OpenAI accounts have a limited quota. If your playbook starts failing with 429 errors, add credit to your OpenAI account or reduce the number of incidents that trigger the playbook.

9.4 Customising the Prompt

The GPT prompt can be tailored to your organisation's needs. Example prompt variations:

- For compliance-focused teams: Add Ask if the incident may involve personal data regulated under GDPR or DPDPA.
- For cloud security teams: Add Identify any affected Azure resources and their risk classification.
- For SOC automation: Add Suggest a Sentinel KQL query to hunt for related activity in the last 7 days.

10 Granting Sentinel Responder Permissions to the Logic App

The Logic App needs the Microsoft Sentinel Responder role so it can post comments back to Sentinel incidents programmatically.

73. In the Azure Portal, navigate to Resource Groups using the left sidebar.
74. Click on the resource group containing your Sentinel workspace (e.g., rg-sentinel-lab).
75. In the resource group menu, click Access control (IAM).
76. Click + Add and select Add role assignment.
77. On the Role tab, search for and select Microsoft Sentinel Responder, then click Next.
78. On the Members tab:
 - Change the Assign access to dropdown from User, group, or service principal to Managed Identity.
 - Click + Select members.
 - In the Managed Identity dropdown, select Logic App.
 - Find and select your Logic App (e.g., GPT-Incident-Responder).
 - Click Select.
79. Click Review + Assign, then Assign.

NOTE Without this role, the Logic App will throw a 403 Forbidden error when attempting to add comments to Sentinel incidents.

The screenshot shows the 'Add role assignment' dialog in the Azure Portal. The 'Members' tab is selected, and the 'Assign access to' dropdown is set to 'Managed identity'. The 'Selected role' is 'Microsoft Sentinel Responder'. The 'Members' section is empty, with a '+ Select members' link. The 'Description' field is optional. On the right, the 'Select managed identities' panel shows a search bar and a list of selected members, including 'gpt-response' with its Object ID. The 'Review + assign' button is at the bottom left, and the 'Select' button is at the bottom right of the right panel.

Home > Resource Manager > Resource groups > mon-checking > Access control (IAM)

Add role assignment

Role: **Members** | Conditions | Review + assign

Selected role: Microsoft Sentinel Responder

Assign access to: ☐ User, group, or service principal ☒ Managed identity

Members: [+ Select members](#)

Name	Object ID	Type
No members selected		

Description: Optional

[Review + assign](#) [Previous](#) [Next](#)

Select managed identities

Some results might be hidden due to your ABAC condition.

Subscription: Azure subscription 1

Managed identity: Logic app (1)

[Search by name](#)

Selected members:

- gpt-response
/subscriptions/c1315d74-155f-4b85-8f65-93172b78fe09/resourceGroups/... [Remove](#)

[Select](#) [Close](#) [Feedback](#)

Create a resource

Home

Dashboard

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FAVORITES

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App Services

Function App

Azure SQL Database

Azure Cosmos DB

Virtual machines

Load balancers

Storage accounts

Virtual networks

Microsoft Entra ID

Monitor

Advisor

Microsoft Defender for Cloud

Home > Resource Manager | Resource groups > mon-checking | Access control (IAM)

Add role assignment

...

X

Role

Members

Conditions

Review + assign

Role

Microsoft Sentinel Responder

Scope

/subscriptions/c1315d74-155f-4b85-8f65-93172b78fe09/resourceGroups/mon-checking

Members

Name	Object ID	Type
gpt-response	d86de4a4-895a-43f6-bdd0-2cf9816b1a0a	Logic app ⓘ

Description

No description

Review + assign

Previous

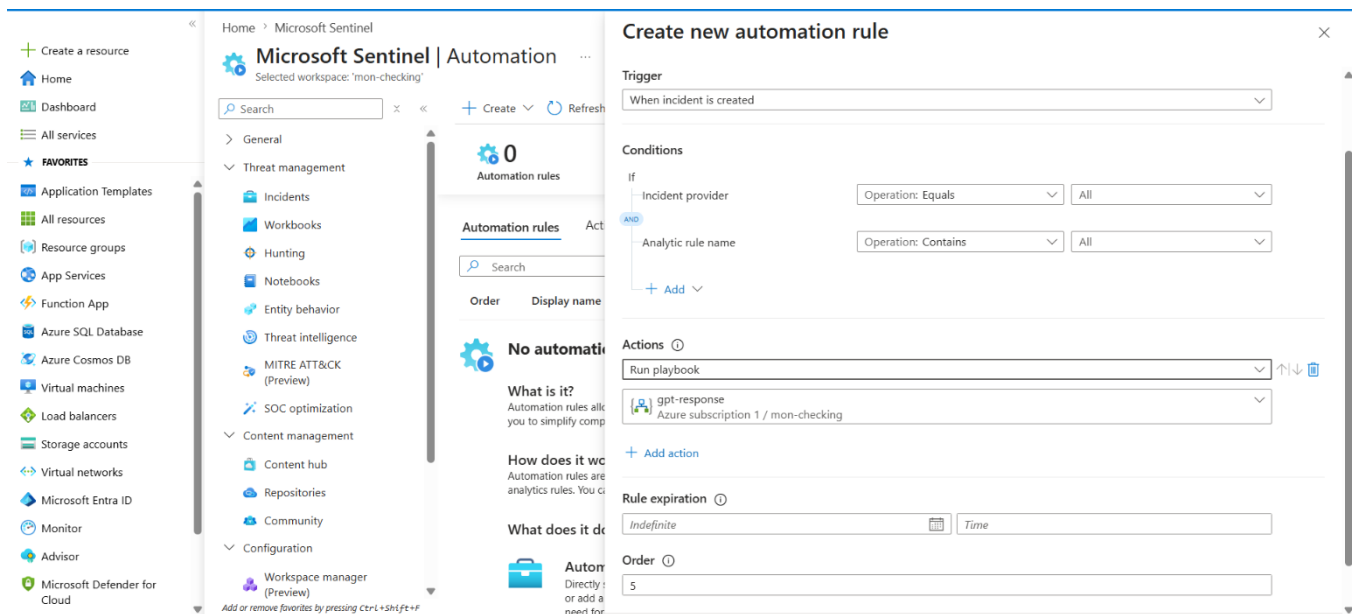
Next

Feedback

11 Automation Rule — Trigger on Incident

An Automation Rule in Microsoft Sentinel automatically executes actions (such as running a playbook) when an incident is created or updated. This wires the GPT playbook to fire every time a new incident appears.

80. In Microsoft Sentinel, click Automation in the left sidebar.
81. Click + Create and select Automation rule.
82. Configure the automation rule:
 - Name: e.g., Trigger GPT Analysis on New Incident
 - Trigger: When incident is created
 - Conditions: (Optional) Add filters such as Severity is High or Medium to limit which incidents trigger the GPT analysis
83. Under Actions, click + Add action.
84. Select Run playbook and choose your Logic App (GPT-Incident-Responder) from the dropdown.
85. Set an Order value of 1 (runs before any other automation actions).
86. Click Apply.



11.1 Testing the Automation

87. In Microsoft Sentinel, click Incidents in the left sidebar.
88. Click + Create incident.
89. Fill in a test incident title such as Test — Tor Login Detected and set Severity to High.
90. Click Save.
91. After a few seconds, click on the incident to open it.
92. Scroll to the Comments section and verify that a comment from the Logic App appears with the GPT-generated analysis.

NOTE If no comment appears within 2 minutes, check the Logic App run history: open the Logic App resource, click Overview, and examine the Runs history for failed executions.

12 Testing the Full Pipeline

After completing all setup steps, run the following end-to-end test to validate that the entire pipeline — from log ingestion to AI-generated incident comment — is functioning correctly.

#	Test Step	Expected Outcome
1	Open Sentinel > Logs and run: SigninLogs take 10	Returns rows — confirms data ingestion is active
2	Open Sentinel > Watchlist > TorExitNodes and confirm the IP list is populated	Rows visible, IPAddress column populated
3	Open Sentinel > Analytics and verify the Tor Exit Node Sign-In Detected rule is enabled	Status shows Enabled
4	Manually create a test incident (Section 11.1)	Incident appears in the Incidents list
5	Open the test incident and check the Comments section	GPT-generated analysis comment is present
6	Open the Logic App > Runs history and inspect the latest run	All steps show green Succeeded status

13 Cost Management & Free Tier Optimisation

Microsoft Azure and OpenAI both charge based on usage. The following practices will help you stay within free-tier limits and avoid unexpected charges.

13.1 Azure Cost Controls

- Set a daily ingestion cap on the Log Analytics Workspace: In the workspace settings, under Usage and estimated costs, set a Daily cap of 10 GB. This is the free-tier daily limit for Microsoft Sentinel.
- Disable unused data connectors: Every active connector that produces data counts toward your ingestion quota. In Sentinel > Data connectors, disconnect any connector whose data you do not actively use in a rule or workbook.
- Set a Budget Alert: In the Azure Portal, go to Cost Management + Billing > Budgets > + Add. Set a monthly budget of \$0 (or a small buffer amount) and configure an alert at 80% and 100% thresholds, sent to your email address.
- Review the Workspace pricing tier: Under the Log Analytics Workspace > Usage and estimated costs, confirm the workspace is on the Pay-As-You-Go (free-tier) pricing and not accidentally provisioned on a Commitment tier.
- Delete the resource group after the project: If this is a lab/demo environment, go to Resource Groups, select rg-sentinel-lab, and click Delete resource group. This removes all billable resources permanently.

13.2 OpenAI API Cost Controls

- Set usage limits in the OpenAI dashboard: Navigate to <https://platform.openai.com/account/limits> and set a hard monthly limit (e.g., \$5) so the API key stops accepting requests when the limit is reached.
- Reduce max_tokens: The GPT call in the Logic App is configured with max_tokens: 500. Reducing this (e.g., to 200) cuts the per-call token cost roughly in half at the expense of a shorter analysis.
- Filter which incidents trigger the GPT playbook: In the Automation Rule (Section 11), add a severity condition of High or Critical only. This prevents low-fidelity, low-priority incidents from consuming API quota.
- Cache or summarise recurring incident types: If you see the same incident type appearing repeatedly (e.g., brute force on the same account), add a logic condition to the Logic App to skip GPT analysis for incidents that have been seen more than N times in 24 hours.

13.3 Free-Tier Limits Summary

Service	Free Allowance	Action if Exceeded
Microsoft Sentinel	10 GB/day log ingestion (first 31 days)	Set daily cap; disable non-essential connectors
Log Analytics Workspace	5 GB/month for free account after trial	Reduce connectors and analytics rule frequency

Azure Logic Apps	4,000 actions/month (Consumption tier)	Add conditions to limit playbook triggers
OpenAI GPT-3.5-Turbo	\$5 free credit (new accounts)	Set hard limit in OpenAI dashboard

14 Troubleshooting Reference

Symptom	Likely Cause	Resolution
Deployment shows Failed	Unsupported data connector licences	Ignore — workspace is still healthy. Proceed with remaining setup steps.
No data in Logs	Data connectors not configured or no data sources	In Sentinel > Data connectors, open each connector and follow the configuration instructions on its blade.
Logic App fails with 403 error	Logic App Managed Identity lacks Sentinel Responder role	Repeat Section 10 — verify the correct Logic App was selected in the IAM assignment.
No GPT comment on incident	Logic App not triggered or HTTP call failing	Check Logic App run history. Verify the API key is correct and begins with Bearer in the Authorization header.
GPT returns empty or truncated response	max_tokens too low or prompt too long	Increase max_tokens to 800 in the HTTP body. Shorten the incident description passed to GPT if it is very long.
UEBA data not appearing	UEBA needs time to build baselines	Wait 24 to 48 hours after enabling UEBA. Ensure Azure AD connector is active.
Unexpected charges on Azure	Daily cap not set or budget alert not configured	Set ingestion cap (Section 13.1). Delete unused connectors and run Delete resource group when the project is complete.

15 Glossary

Term	Definition
Analytics Rule	A scheduled or real-time KQL query in Sentinel that generates alerts when matching events are found.
ARM Template	Azure Resource Manager template — an infrastructure-as-code JSON file used to provision Azure resources declaratively.
Automation Rule	A Sentinel feature that triggers actions (such as running a playbook) automatically when an incident meets defined criteria.
Data Connector	A pre-built integration that streams logs from an external source (e.g., Azure AD, Syslog, CEF) into the Log Analytics Workspace.
Diagnostic Settings	Azure configuration that routes platform metrics and logs from a resource to a Log Analytics Workspace or Storage Account.
GPT	Generative Pre-trained Transformer — the AI language model from OpenAI used in this project to generate natural-language incident analyses.
IAM	Identity and Access Management — the Azure system for assigning roles and permissions to users, groups, and managed identities.
Incident	A grouped set of related alerts in Microsoft Sentinel that represents a potential security event requiring investigation.
KQL	Kusto Query Language — the query language used to analyse log data in Azure Log Analytics and Microsoft Sentinel.
Logic App	An Azure serverless workflow engine used to build and execute automated playbooks without writing traditional application code.
Managed Identity	An Azure AD identity automatically managed by Azure, used by services such as Logic Apps to authenticate to other Azure resources without stored credentials.
Playbook	In Microsoft Sentinel, a playbook is a Logic App workflow that automates a security response action when triggered by a Sentinel automation rule.
Resource Group	An Azure container that holds related resources for a project, enabling collective management, permissions, and billing.
SIEM	Security Information and Event Management — a platform that collects, correlates, and analyses security log data to detect threats.
SOAR	Security Orchestration, Automation, and Response — the capability to automate security response workflows, represented in this project by Azure Logic Apps.
Tor Exit Node	An IP address used as the final relay in the Tor anonymisation network. Traffic from these nodes may indicate an attacker trying to hide their origin.

UEBA	User and Entity Behaviour Analytics — an ML-based Sentinel feature that detects anomalous behaviour by comparing current activity to established baselines.
Watchlist	A lookup table imported into Microsoft Sentinel (typically as a CSV) and queryable via KQL to enrich alerts with external threat intelligence.

End of Document